

Milestone 1 Proposal: Identifying Risk Factors for Postpartum Depression

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Topic

Identifying Risk Factors for Postpartum Depression. This project will analyze maternal, socioeconomic, family, obstetric, and neonatal characteristics to determine which factors are most strongly associated with an elevated risk of postpartum depression.

Business Problem

Postpartum depression can affect maternal health, family functioning, and the well-being of infants, making early identification an important maternal health concern. The Centers for Disease Control and Prevention reports that approximately one in eight women with a recent live birth report symptoms of postpartum depression. Identifying elevated risk earlier may help healthcare providers focus screening, education, and support on patients who may benefit from additional follow-up.

This project will explore several research questions: Which maternal, socioeconomic, family, pregnancy, and neonatal characteristics are most strongly associated with postpartum depression risk? Can postpartum depression risk be predicted using information that would reasonably be available before or shortly after delivery? How accurately can statistical and machine learning models distinguish between women at lower and higher risk? The analysis will also examine whether certain categories of risk factors, such as previous mental health history, social support, pregnancy complications, or socioeconomic conditions, appear to be more influential than others.

Datasets

The primary dataset will be the **Data for Postpartum Depression Prediction in Bangladesh** dataset published through Mendeley Data. The current dataset contains information

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from 800 postpartum women who had given birth within the previous 24 months. It includes sociodemographic characteristics such as age, education, residence, employment, and income; family characteristics such as household structure and relationships; personal and medical history; pregnancy and delivery information; neonatal health characteristics; and postpartum experiences.

The dataset also contains responses and scores from established depression screening measures, including the Edinburgh Postnatal Depression Scale (EPDS), Patient Health Questionnaire-2 (PHQ-2), and Patient Health Questionnaire-9 (PHQ-9). Depression screening results will be used to define the outcome variable rather than as predictor variables. Individual depression-screening questions and other variables that directly reveal the outcome will be excluded from predictive features to avoid data leakage.

CDC Pregnancy Risk Assessment Monitoring System research and other maternal mental health literature will be used as supporting evidence to compare the findings with established postpartum depression patterns.

Methods

The project will begin with data inspection and cleaning, including evaluating missing values, duplicate observations, categorical variables, class balance, and potentially inconsistent responses. A data dictionary will be created to document the variables retained for analysis and distinguish predictors from depression-screening outcome variables.

Exploratory data analysis will be used to examine the distribution of postpartum depression risk across demographic, socioeconomic, family, medical, pregnancy, and neonatal characteristics. Comparative visualizations and statistical tests will be used to identify potentially meaningful relationships between these characteristics and postpartum depression risk.

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A logistic regression model will provide an interpretable baseline for determining the direction and relative strength of associations between selected predictors and postpartum depression risk. A tree-based classification model, such as Random Forest, will also be evaluated to determine whether nonlinear relationships and interactions improve predictive performance. Model performance will be assessed using measures such as recall, precision, F1 score, ROC-AUC, and the confusion matrix.

Feature importance and model coefficients will be examined to determine which variables contribute most strongly to prediction. Particular attention will be given to avoiding target leakage by excluding EPDS, PHQ-2, PHQ-9, and other symptom variables that directly define or closely approximate the target outcome.

Ethical Considerations

Postpartum depression is a sensitive mental health issue, and predictive analysis creates several ethical concerns. A statistical model should not be interpreted as diagnosing postpartum depression or determining whether an individual requires treatment. False-negative predictions could fail to identify someone who needs additional support, while false-positive predictions could unnecessarily label an individual as high risk.

The dataset contains sensitive characteristics involving mental health history, abuse, family relationships, socioeconomic conditions, and pregnancy experiences. Results must be presented in a manner that avoids blaming individuals or demographic groups for differences in postpartum depression risk. Associations involving income, education, marital status, residence, or other characteristics should be interpreted within their broader social and healthcare context.

Cultural context is another important consideration. Because the primary dataset was collected in Bangladesh, relationships identified in the analysis may not apply to postpartum

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populations in the United States or other countries. The model will therefore be presented as an exploratory risk-factor model based on the population represented in the dataset rather than as a universal clinical prediction tool.

Challenges & Issues

One challenge will be preventing data leakage. The dataset contains several depression screening measures and individual questionnaire responses that could make prediction artificially easy if variables used to define postpartum depression were also included as predictors. These variables will need to be carefully identified and removed from the predictive feature set.

Another challenge may involve categorical variables with many response levels, missing observations, and imbalanced outcome classes. Appropriate encoding, preprocessing, and evaluation methods will be required to prevent these issues from distorting model performance.

The sample consists of 800 women from Bangladesh, which limits the generalizability of the findings. Cultural, economic, healthcare, and social differences may affect both postpartum depression and the factors associated with it. The cross-sectional and self-reported nature of many variables will also limit the ability to establish causal relationships. The final analysis will therefore focus on associations and predictive relationships rather than claims of causation.

References

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